

Introduction to Blockchain and Smart Contracts

Lecture 14: Side Chains & Rollups

Subodh Sharma | IIT Delhi | March 19, 2026



Sidechains, Statechains, L2 Rollups

Ethereum is SLOW

- **Sidechain:** Move Txns to separate chain and occasionally connect back to Ethereum
 - Have their own validators and security/consensus measures
 - It has its own block production rules and its own bridge to move assets in or out
- **State channels & Rollups:** Keep the activity off-chain between a small set of participants and settlements on Ethereum

Sidechains

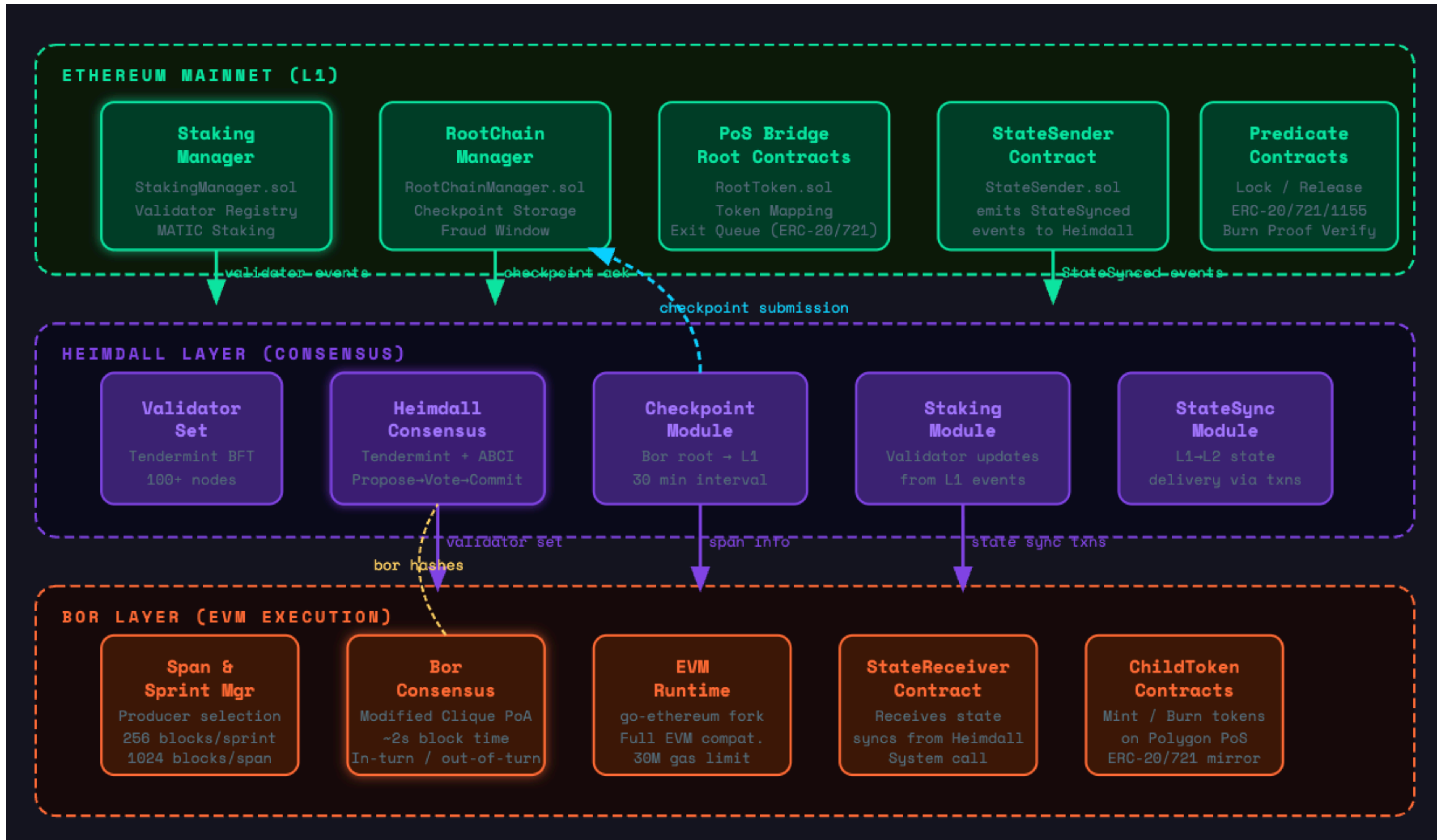
- How do they work? Examples include: Polygon PoS [Used by Nike, Starbucks, ..]
 - Anita locks 10 ETH in a **bridge contract** on Ethereum
 - The sidechain bridge mints or releases 10 “**wrapped ETH**” for Anita on the side chain
 - Anita now transacts on the sidechain with cheaper costs
 - She burns/releases the wrapped asset on the side chain
 - The bridge unlocks the original ETH on Ethereum
 - Of course accounting for Bridge fees, de-pegging of wrapped assets, transaction fee on the main-net, etc.

Polygon PoS — Sidechain

- Two layered architecture:
 - **Bor layer:** The execution/block production layer
 - **Heimdall layer:** The consensus/validator layer; uses **Tendermint BFT**
- The Ethereum layer is used to host staking contracts where Polygon checkpoints are posted

Polygon PoS — Sidechain

Full Architecture



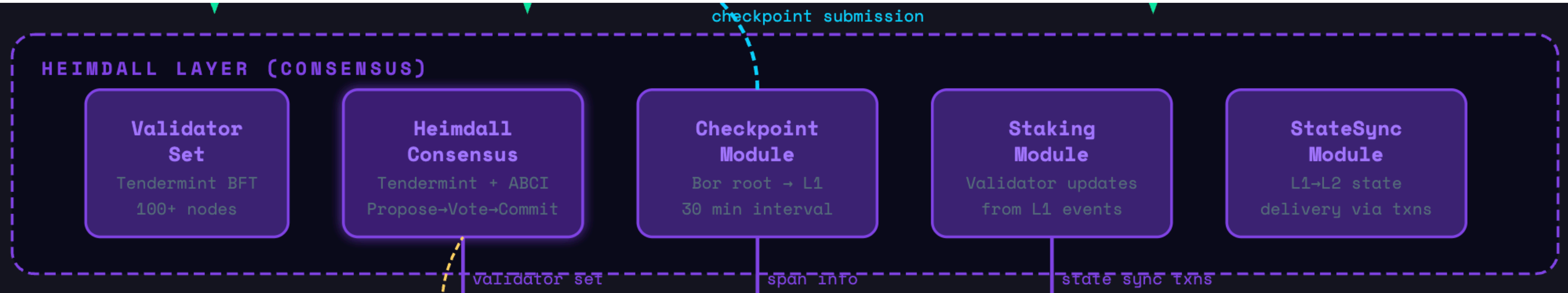
How do the two layers interact?

- Bor produces Blocks
- Heimdall validators collect Bor block hashes compute Merkle root
- Proposer submits MsgCheckPoint to Heimdall Pool
- Heimdall Tendermint voting (2/3 +1 validators sign off)
- Checkpoint block is committed

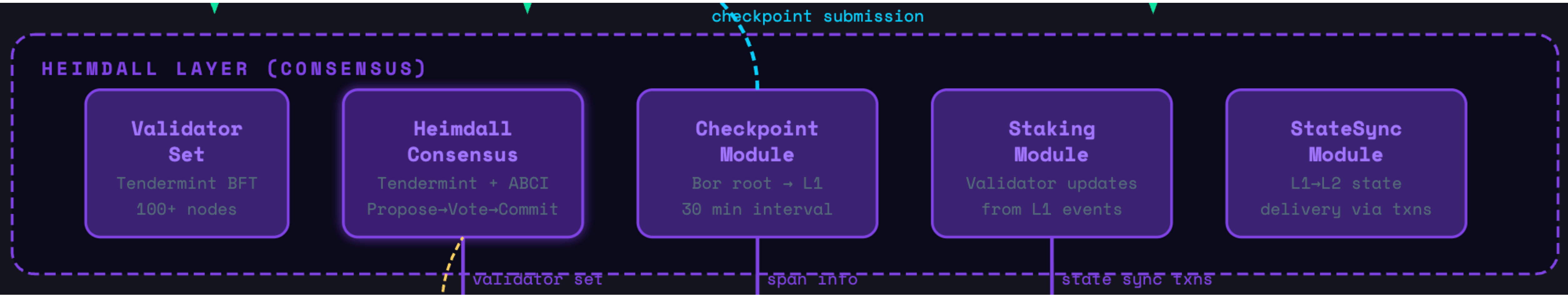
- L1 submission: Proposer sends the txn to RootChainManager on Ethereum
- Finality: After L1 confirmation, Bor range is officially cryptographically anchored.



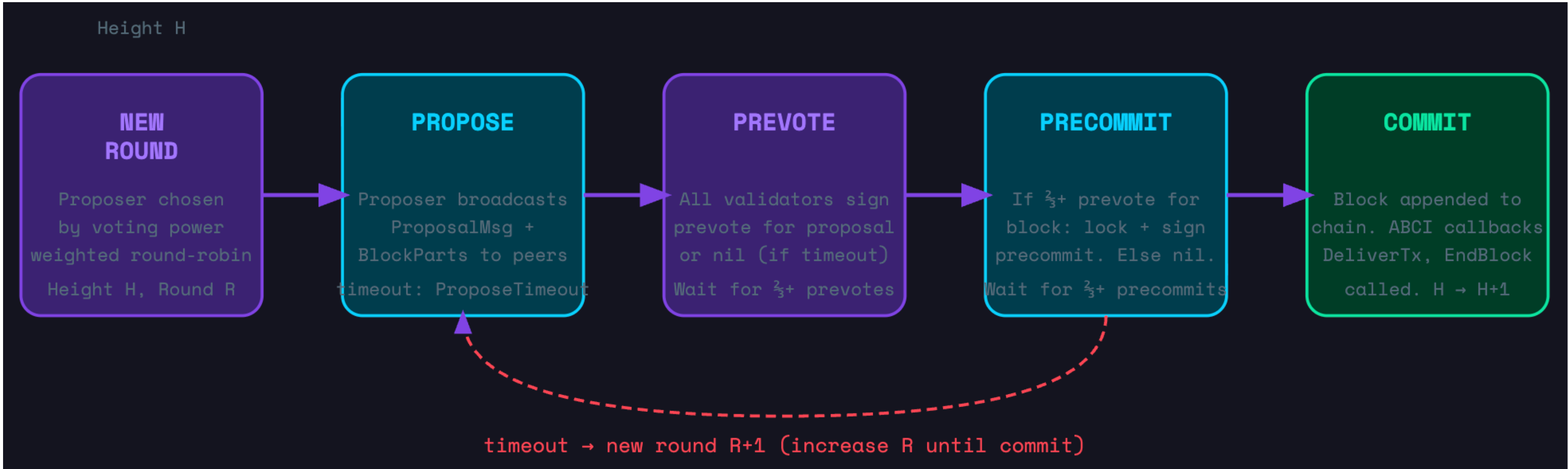
- **Bor** is the execution client of Polygon PoS
 - Consists of a set of nodes that produce blocks regularly
 - Accepts txns from users; orders them into blocks; execute EVM logic; updates accounts balances and contract state; exposes Eth RPC interface to developers
 - Based on Go Ethereum (or **Geth**)
 - **Sprint-Span Framework:**
 - **Sprint:** Assigned to one primary producer. Duration is 32 sec to produce 16 blocks
 - Per block ~2 sec — **How?**
 - **Span:** A validator subset is assigned for 6400 blocks. Duration is 3.5 hrs



- **Heimdall Layer:**
 - **Checkpoint:** handles the proposal, voting and submission of Bor block range commitments to Ethereum
 - **Staking:** Manages actives validator set and stake weights
 - **StateSync:** Records state-sync methods emitted by L1 (StateSender Contract)
 - Bor picks these up at sprint boundaries



- **Heimdall Layer:**
 - **Consensus:** Uses Tendermint (now known as CometBFT)
 - BFT up to 1/3 of the total stake controlled by malicious validators



L2 Rollups

- Design Goal (in the context of Ethereum):
 - Keep Ethereum as a trust anchor
 - Move most of execution of L1 — **saves cost**
 - Periodically commit results back to Ethereum
 - Ethereum enforce correctness and/or data availability
- What exactly is it?
 - Executes txns off-chain or off-L1; Batches txns together
 - Post compressed transaction data or state commitments to Ethereum
 - Final settlement is conducted on Ethereum L1

L2 Rollups

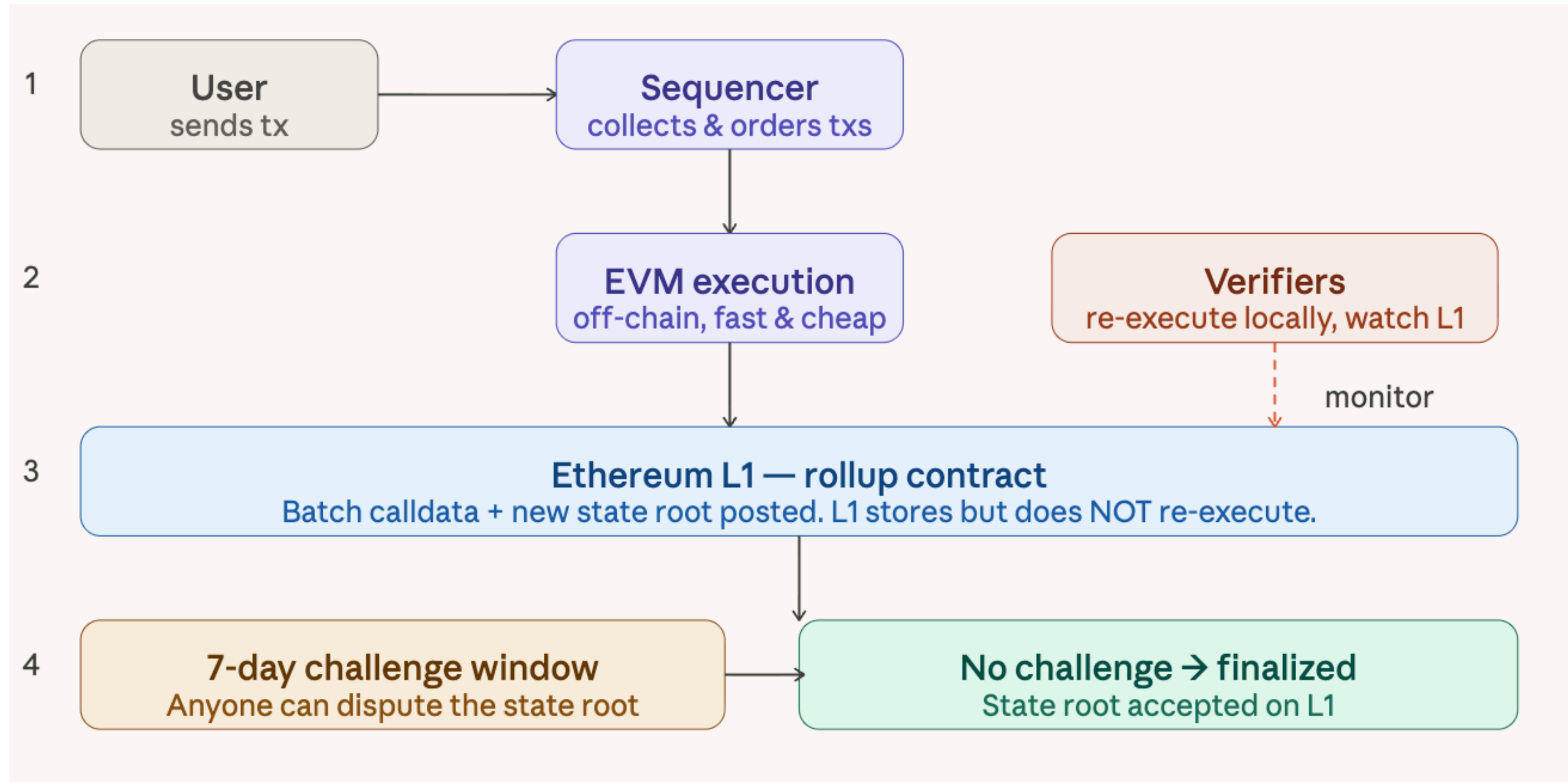
Core Architecture

- On-chain (ie L1) contracts
 - for tracking deposits from L1 to L2 and to store commitments about L2 state
 - to verify proofs or handle disputes and finalise withdrawals back to L1
- Off-chain execution
 - User sends txns to a **rollup**, which executes the txns, updates the local state and periodically posts batch data and commitments to L1
- Actors: User, Sequencer/Operator, Proposer, Verifier/Challenger/Prover

L2 Rollups

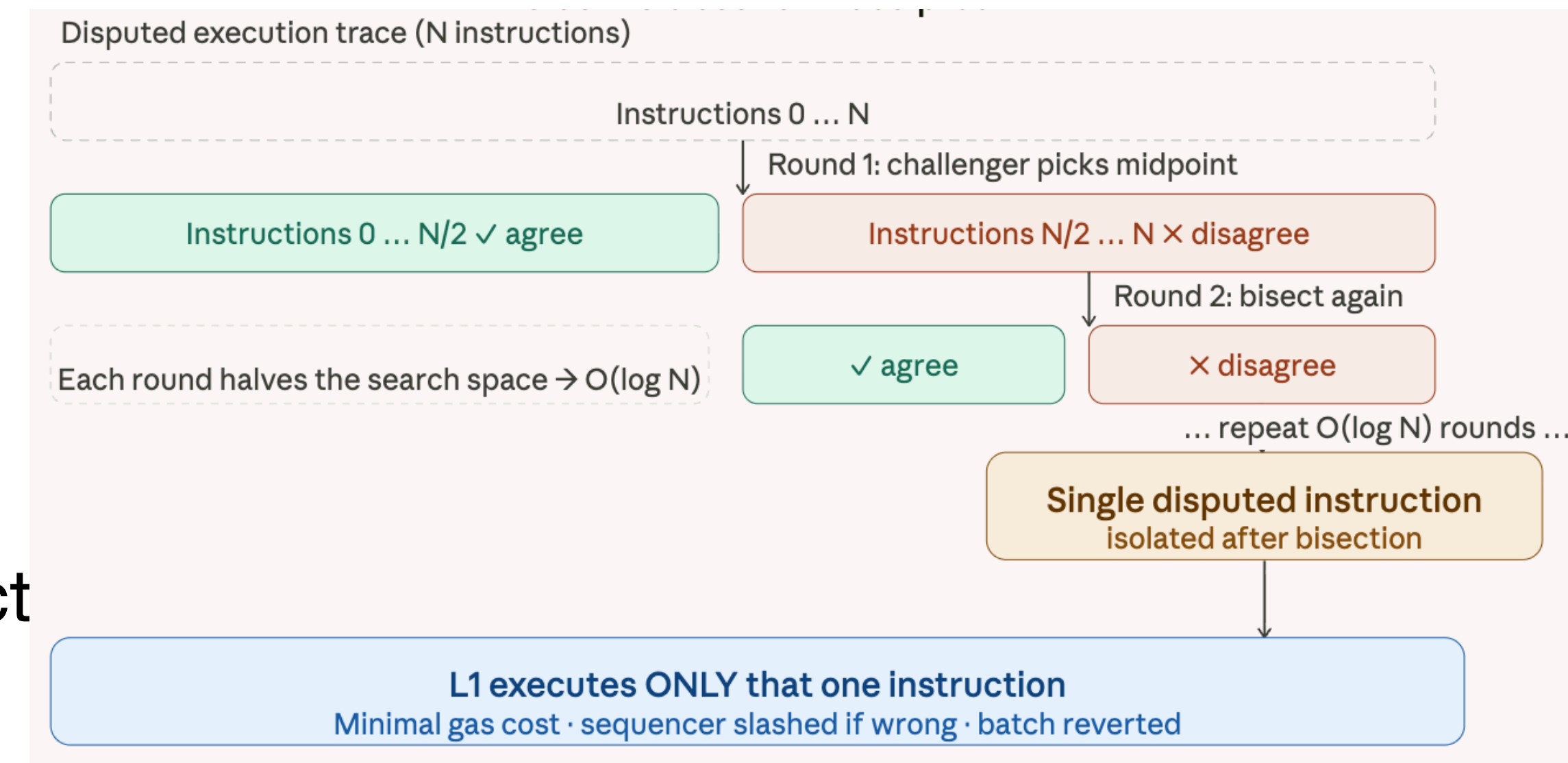
Optimistic Rollups

- Assume batched txns are correct unless disputed



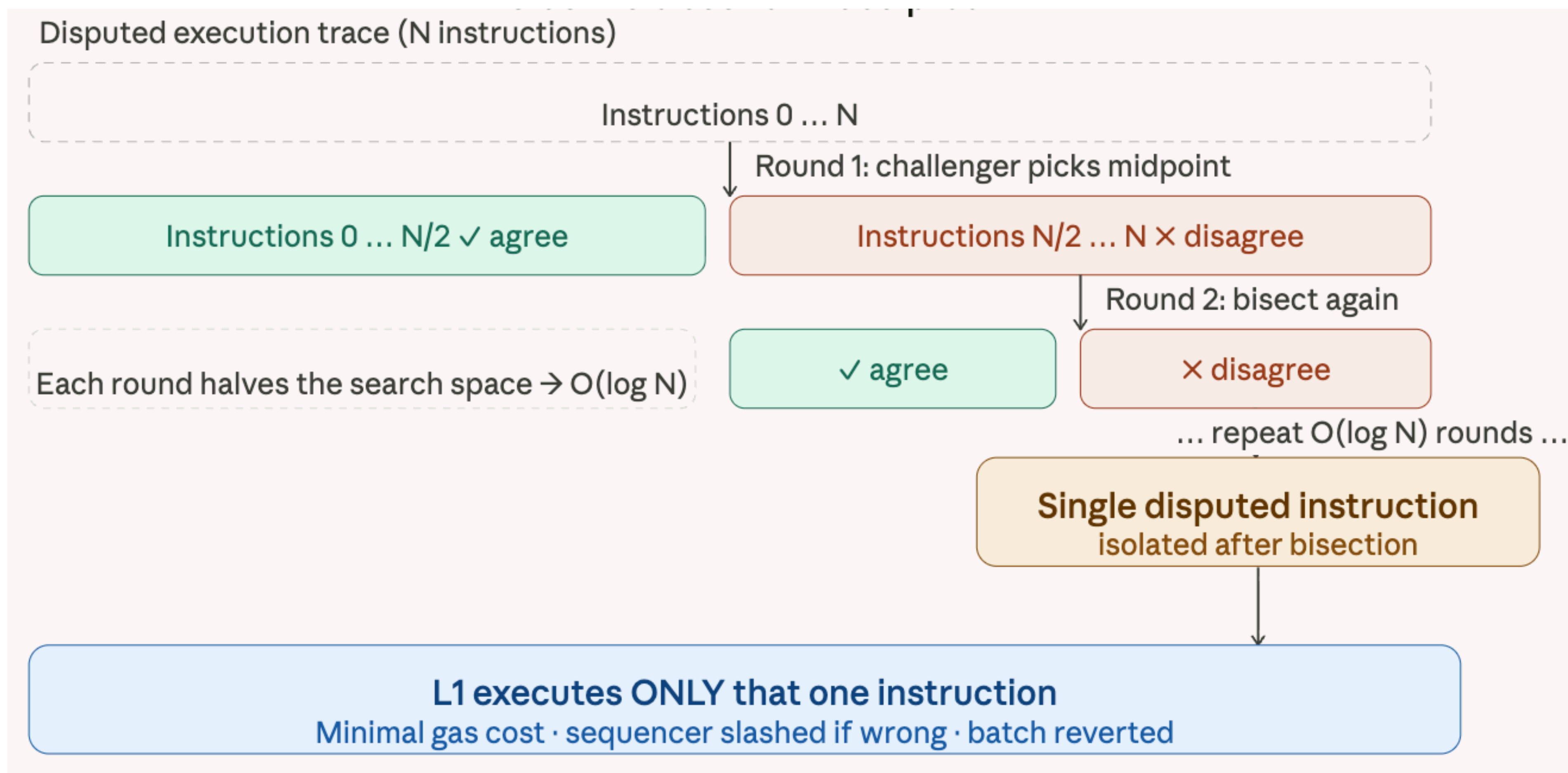
Fraud Proof

- Purpose:
 - Someone claims the L2 state transition is correct
 - Someone else challenges that claim
 - A dispute game determines whether the claim is valid
- Two approaches:
 - **Single round:** The challenger submits the full re-execution of the disputed txn batch to L1 —> L1 re-runs it and validates the claims —> simple and expensive
 - **Multi-round interactive:** Identify the offending txn —> L1 executes that offending txn —> complex but efficient on gas



Fraud Proof

Multi-round Interactive Bisection Proof



Data Availability

- All the txn data goes to L1
- Moving these assets from L2 to L1 goes through a **bridge contract**
 - Hosted on L1
- L1 → L2: Depositing from L1 to L2 is quick.
 - Lock ETH or tokens in the bridge contract → sequencer sees the deposit events and credits your L2 account
- L2 → L1 (withdrawing): This is slow. You initiate a withdrawal from L2; the sequencer includes it in the batch; gets posted in L1; and then wait for 7 days; post the challenge window finalise the withdrawal and claim the funds

L2 Optimistic Rollups

Security Model

- 1-of-N honest verifier security model
- The sequencer must post a **bond** (a stake) that gets slashed if a fraud proof succeeds
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